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### Finger Massage Apparatus

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#### Abstract

A finger massage apparatus for calming and relaxation wherein a bead, wheel, barbell, or other tactile feature within a channel may be used to provide a tactile or massaging sensation to a user. A user may place their finger in the device, or wear the device normally as with jewelry, and then manipulate the bead, wheel, or barbell to massage the finger. The device may be used in conjunction with breathing, focus, or meditation exercises and may be used to help focus the user on those calming activities. The device may include a number of variations, including those based on a ring with a platform, a sleeve, a double-ring structure, or other configurations to provide a variety of wearable devices with adaptable tactile sensations.

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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application claims priority under 35 U.S.C. § 119 (e) from U.S. Provisional Application Ser. No. 63/558,057, filed Feb. 26, 2024, entitled “Deep Breathing Fidget Ring,” the disclosure of which is hereby incorporated by reference herein in its entirety.

### FIELD OF THE INVENTION

[0002] The present invention relates generally to the field of health and wellness accessories. More specifically, the invention is in the subfield of massage and tactile sensation devices.

### BACKGROUND OF THE INVENTION

[0003] Many individuals suffer from anxiety, stress, nervousness, or other mental and physical symptoms of distress. There are a number of methods for addressing these symptoms, including, but not limited to, pharmaceuticals, breathing methods, meditation, and devices such as sound machines or other devices that allow a person to focus or otherwise channel their nervous energy to remain or become calm. However, most of these methods for addressing nervous energy, anxiety, and stress require harsh drugs that require a prescription, or breathing and meditation methods that are difficult to focus on, particularly when an individual may be in a state of mental distress. Furthermore, an individual may not be in a place where it is possible to fully relax or to use a sound machine or other aids to help in focusing and maintaining a breathing or meditation exercise.

[0004] As such, there is a need in the art for a simple, effective, and portable device that may help an individual to relax, calm themselves, or participate in methods for reducing stress, anxiety, and nervousness.

### SUMMARY OF THE INVENTION

[0005] An aspect of an embodiment of the present invention provides a finger massage apparatus that allows a user to place the device on their finger and then manipulate a bead, wheel, barbell, or other feature to give the user a tactile sensation. The user may manipulate the bead, wheel, barbell, or other feature with another finger on the hand in which the device is mounted, or with their opposing hand. The user may then experience both a direct tactile sensation from the purposeful manipulation of the device and an indirect tactile sensation from the bead, wheel, barbell, or other feature engaging with the user's finger within the finger massage apparatus. The device may be used in conjunction with breathing exercises, with meditation exercises, and to provide a tactile sensation for the user to focus on. The user may also gain some relaxation benefit because the device allows them a small item to play with, manipulate, or fidget. The tactile and massaging function of the present invention also serves to stimulate blood flow, reduce arthritic effects, and may stimulate the vagus nerve, providing additional benefits to the user.

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## Description

### DESCRIPTION OF THE DRAWINGS

[0006] The accompanying drawings, which are incorporated into and form a part of the instant specification, illustrate several aspects and embodiments of the present invention and, together with the description herein, serve to explain the principles of the invention. The drawings are provided only for the purpose of illustrating select embodiments of the invention and are not to be construed as limiting the invention.

[0007] FIGS. 1-3 provide schematic depictions of an exemplary embodiment of a finger massage

apparatus.

[0008] FIG. **4** provides a schematic depiction of an exemplary embodiment of a finger massage apparatus with a detachable platform.

[0009] FIGS. **5** and **6** provide schematic depictions of an exemplary embodiment of a finger massage apparatus with a cuff-like ring.

[0010] FIG. **7** provides a schematic depiction of an exemplary embodiment of a finger massage apparatus with a size-adjustable ring.

[0011] FIGS. **8** and **9** provide schematic depictions of an exemplary embodiment of a finger massage apparatus with a spherical bead.

[0012] FIG. **10** provides a schematic depiction of an exemplary embodiment of a finger massage apparatus with a transverse channel.

[0013] FIG. **11** provides a schematic depiction of an exemplary embodiment of a finger massage apparatus with two transverse channels.

[0014] FIGS. **12** and **13** provide schematic depictions of an exemplary embodiment of a finger massage apparatus with a sleeve.

[0015] FIGS. **14** and **15** provide schematic depictions of an exemplary embodiment of a finger massage apparatus with a sleeve and a sheath for containing a bead.

[0016] FIGS. **16-18** provide schematic depictions of exemplary embodiments of a finger massage apparatus with a sleeve in different lengths.

[0017] FIG. **19** provides a schematic depiction of an exemplary embodiment of a finger massage apparatus with two channels.

[0018] FIGS. **20** and **21** provide schematic depictions of an exemplary embodiment of a finger massage apparatus with two rings.

[0019] FIG. **22** provides a schematic depiction of an exemplary embodiment of a clamp style finger massage apparatus.

[0020] FIGS. **23A-23F** provide schematic depictions of exemplary embodiments of different channel designs for the platform of a finger massage apparatus.

[0021] FIGS. **24A-24D** provide schematic depictions of exemplary embodiments of platforms with channel detents for a finger massage apparatus.

[0022] FIGS. **25A-25F** provide schematic depictions of the engagement of a bead, roller, or wheel with the channel of the platform of exemplary embodiments of a finger massage apparatus.

[0023] FIGS. **26A-26D** provide schematic depictions of exemplary embodiments of beads for a finger massage apparatus.

[0024] FIGS. **27A-27D** provide schematic depictions of exemplary embodiments of a roller wheel for a finger massage apparatus.

[0025] FIG. **28** provides a schematic depiction of an exemplary embodiment of a split bead for a finger massage apparatus.

#### DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS OF THE INVENTION

[0026] FIGS. **1-3** provide depictions of an exemplary embodiment of a finger massage apparatus **10**, which may comprise a platform **12** with a ring **18** affixed or otherwise attached to the bottom surface of the platform **12**. The platform **12** may also include or incorporate a channel **14** and a bead **16** disposed within the channel **14**. The bead **16** may itself comprise an upper contact surface **20** and a lower contact surface **21**, with one or more bead channels **22** disposed between the upper contact surface **20** and the lower contact surface **21**. The bead channels **22** are adapted to mate with the channel **14** to allow the bead **16** to translate, slide, or move through the channel **16** when manipulated by a user. It should be appreciated that the ring **18** of the finger massage apparatus **10** may not be completely circular and may incorporate a gap or aperture where the ring **18** contacts the platform **12** of the finger massage apparatus **10** in order to allow the lower contact surface **21** of the bead **16** to translate or move fully through the length of the channel **14**.

[0027] The upper contact surface **20** of the bead **16** extends above the upper surface of the platform

12. Similarly, the lower contact surface **21** of the bead **16** extends below the lower surface of the platform **12**. When a user inserts his or her finger into the ring **18** of the finger massage apparatus **10**, the lower contact surface **21** of the bead **16** will be positioned to contact the user's finger while the upper contact surface **20** of the bead **16** will be exposed above the platform **12** such that the user may manipulate the bead **16** and move or displace it through the channel **14**. When the user moves or displaces the bead **16** through the channel **14**, the user will experience a tactile sensation or massage on the finger which is inserted into the finger massage apparatus **10**. Furthermore, the user will experience two types of tactile sensation. Firstly, the user will feel a direct, or active, tactile sensation when the user contacts the upper contact surface **20** of the bead **16** to move or displace the bead **16**. Secondly, the user will feel an indirect, or passive, tactile sensation when the lower contact surface **21** of the bead **16** is moved or displaced and contacts the user's finger that is within the ring **18** of the finger massage apparatus **10**.

[0028] Still referring to FIGS. 1-3, a user may manipulate or use the finger massage apparatus **10** in a number of ways to provide a calming, relaxing, or anxiety-reducing experience. For example, the user may place the finger massage apparatus **10** on their finger with the platform **12** facing upwards as with a traditional ring and use their other hand or fingers to manipulate the bead **16**. Similarly, a user may place the platform **12** on the underside, or palm side of their finger and use either a finger from the opposing hand or their thumb to manipulate the bead **16**. The user may also use the finger massage apparatus **10** in conjunction with breathing exercises, mantras, meditation, calming music, essential oils, or other calming methods or devices to relax or relieve anxiety. For example, a user may time the manipulation of the bead **16** with their breathing by extending the bead **16** away from them in the channel **14** on inhale and retracting the bead **16** towards them during exhale. However, the user may manipulate the bead **16** and move it through the channel **14** in any way or in any arrangement as desired or required for a particular breathing exercise, meditation exercise, or other calming method. It should be appreciated that the finger massage apparatus **10** may also provide a fidget device to help calm or relax someone with nervous energy or anxiety issues.

[0029] FIG. 4 provides an exploded view of a finger massage apparatus **10** with a separable platform **12** and ring **18**. The platform **12** of the finger massage apparatus **10** may include or incorporate a channel **14** and a bead **16** disposed with and in slidable communication with the channel **14**. The ring **18** may include or comprise one or more ring pads **24**. Similarly, the platform **12** may include or comprise a platform recess **26** that is adapted or otherwise configured to mate with the ring pads **24** such that the platform **12** may be removed and reattached to the ring **18**. A user may then replace the platform **12** for another platform **12** that may include a differently shaped bead **16** or channel **14** for a different tactile sensation, or for another platform **12** which has a different aesthetic style or design. In this way, a single ring **18** may allow the user to experience multiple physical sensations or to match with different types of outfits, situations, or styles. It should be appreciated that any number of attachment means may be used between the platform **12** and ring **18**. For example, the platform **12** may be affixed to the ring **18** via a snap mechanism, a sliding mechanism incorporating dovetails or other geometric features to secure the platform **12** to the ring **18**, magnets, fasteners, pins, or any other attachment mechanism that is suitable for a particular application of the finger massage apparatus **10**.

[0030] FIGS. 5 and 6 provide perspective and cross-sectional views, respectively, of an exemplary embodiment of a finger massage apparatus **10** with cuff horns **28** in place of a ring. As before, the finger massage apparatus **10** may comprise a platform **12** with a channel **14** adapted to accept or otherwise carry a bead **16**. The bead **16** may also comprise an upper contact surface **20**, lower contact surface **21**, and one or more bead channels **22** configured or adapted to mate with or engage the channel **14** of the platform **12**. The cuff horns **28** may be made of a flexible, bendable, or springy material such that the finger massage apparatus **10** may be provided with cuff horns **28** sized to fit or grip a relatively small finger while allowing the cuff horns **28** to flex to expand and fit a larger finger without undue discomfort. This expansion and flex of the cuff horns **28** may then

allow one finger massage apparatus **10** to comfortably fit a wider variety of finger sizes while providing adequate grip on the user's finger to allow for manipulation of the bead **16** without displacing the finger massage apparatus **10**. It should be appreciated that the cuff horns **28**, or the ring **18** as shown in other exemplary embodiments of the finger massage apparatus **10**, may include an inner lining, material, or texture to provide additional grip of the finger massage apparatus **10** to the user's finger. This additional grip may facilitate manipulation of the bead **16** during breathing or other calming exercises without slipping or displacement of the finger massage apparatus **10**.

[0031] FIG. 7 provides a schematic depiction of an exemplary embodiment of the finger massage apparatus **10** with a user-adjustable sizing mechanism. The finger massage apparatus may include or comprise a platform **12** with a channel **14**, and a bead **16** moveably disposed within the channel **14**. A ring **18** may be affixed or otherwise attached to the platform **12** and comprise a sizing aperture **35**. The finger massage apparatus **10** may also comprise a sizing band **30** in communication with the underside of the platform **12**, the inner perimeter of the ring **18**, or both. The sizing band **30**, which may be made of a flexible material, may also comprise one or more notches **32** or ridges and a pull tab **34** extending through the sizing aperture **35**.

[0032] To size the ring **18** of the finger massage apparatus **10**, a user may place the ring **18** on their finger with the sizing band **30** in a slack or relaxed position. The user may then pull on the pull tab **34**, pulling more length of the sizing band **30** through the sizing aperture **35**, decreasing the effective internal circumference of the sizing band **30**, and consequently providing a tighter fit to the user's finger. The notches **32** or ridges along the length of the sizing band **30** may then interface or otherwise interact with the edge of the sizing aperture **35** or other features on the ring **18** to provide a ratcheting function wherein the sizing band **30** will maintain tension and sizing until the user releases the sizing band **30** by manipulating the pull tab **34**.

[0033] FIGS. 8 and 9 provide a schematic depiction of an exemplary embodiment of a finger massage apparatus **10** and a sectional view of the bead **16** and platform **12**, respectively. As shown, the finger massage apparatus **10** may comprise a platform **12** in communication with a ring **18**. The platform **12** may comprise a circular structure with a curved cross section to cradle a spherical bead **16**. One or more retaining protrusions **36** may extend from the platform **12** to make contact with and interface with the bead **16**. It should be appreciated that the retaining protrusions **36** may comprise prongs or other protrusions which grip or interact with the surface of the bead **16**, or that the retaining protrusions **36** may interface with a groove or other geometric feature on the bead **16** to retain the bead **16** within the platform **12**. Furthermore, it should be appreciated that the spherical bead **16** may roll in one or more directions, allowing for a different tactile sensation for both direct and indirect tactile sensations.

[0034] FIGS. 10 and 11 provide depictions of exemplary embodiments of a finger massage apparatus **10** with circumferential channels **14**. Referring to FIG. 10, the finger massage apparatus **10** may comprise a ring **18** with a channel **14** and a bead **16** moveably disposed within the channel **14**. The channel **14** may be disposed such that the bead **16** moves circumferentially around the ring **18** to provide a different direction of massage compared to a channel **14** which is disposed parallel to, or in a plane parallel to, the central axis of the ring **18** (as shown in FIGS. 1-7). Referring to FIG. 11, the ring **18** may include a first channel **14** and a second channel **15** with a bead **16** disposed within each of the first channel **14** and second channel **15**. As shown, both the first channel **14** and second channel **15** may be orientated to guide the bead **16** along the circumference of the ring **18** to provide a different tactile sensation to the user. In certain embodiments, as shown, the bead **16** may take on a different shape to provide a different tactile sensation. For example, the bead may be flat with a curved upper surface, may have a spherical or hemispherical upper surface, or may have an upper surface that is pointed, flattened, or in any shape as desired or required for a particular application.

[0035] FIGS. 12 and 13 provide depictions of exemplary embodiments of a finger massage apparatus **10** having a sleeve **38** for enclosing or attaching to a user's finger. The sleeve **38** may be

in communication with a frame **40**, which may enclose a frame channel **42**. A bead **16** may be disposed within the frame channel **42** and may slide, roll, or both, along the length of the frame channel **42**. It should be appreciated that the sleeve **38** may be made of a rigid or flexible material. In the case of the flexible material, the sleeve **38** may be sized such that the sleeve **38** expands to fit snugly around the user's finger while allowing the user to flex or move their finger freely. In certain embodiments, the frame **40** may be molded directly into the material of the sleeve **38**, or the frame **40** may be a separate piece that is affixed or otherwise attached to the sleeve **38** through any means including, but not limited to, adhesives, fasteners, friction welding, or any other attachment means as desired or required for a particular application. Furthermore, it should be appreciated that the bead **16** may extend above the outer surface of the sleeve **38** and below the inner surface of the sleeve **38** such that a user may manipulate the bead **16** while it is in contact with the user's finger in the sleeve **38**.

[0036] FIGS. **14** and **15** provide depictions of an exemplary embodiment of a finger massage apparatus **10** with a sleeve **38**. A bead **16** may be held in place by a sheath **44** of flexible material. The sheath **44** may be shaped or otherwise configured into an elongated or other shape to allow the bead **16** to move within the sheath **44** and provide a massaging function similar to embodiments of the finger massage apparatus **10** described with reference to other figures herein. For example, the sheath **44** may be rectangular or oval in shape, and may be a covering molded into, adhesively attached, or otherwise attached to a sleeve **38** made of a flexible material to entrain the bead **16** within the space between the sleeve **38** and sheath **44**. The bead **16** may then slide fore and aft within the space between the sleeve **38** and sheath **44** to provide a massaging function. In certain embodiments, the sleeve **38** may be made of either a flexible or rigid material and may contain an aperture that is covered both on the inner surface of the sleeve **38** and the outer surface of the sleeve **38** by the sheath **44**. In such embodiments, the sheath **44** may act like a bag or enclosure to entrain the bead **16** and allow the bead **16** to be moved within the space enclosed by the sheath **44** to provide a massaging function.

[0037] It should be appreciated that the finger massage apparatus **10** may include or comprise any number of materials or features. For example, the bead **16** may take on any number of shapes, sizes, or geometries, and the sleeve **38** may comprise any materials, including both rigid and flexible materials. Furthermore, the sleeve **38** may include materials of different colors, transparent or opaque materials, or may serve as the background or substrate for designs or other visual flourishes.

[0038] FIGS. **16-18** provide depictions of exemplary embodiments of a finger massage apparatus **10** with a sleeve **38**. The finger massage apparatus **10** may include any number of variations or alterations to its geometry and features to be adapted for a particular use or application. For example, referring to FIGS. **16** and **17**, the sleeve **38** may be longer such as to enclose the user's finger across more than one knuckle. In certain embodiments, the sleeve **38** may be long enough to encompass the proximal and medial phalanges, or the proximal, medial, and distal phalanges. Similarly, the finger massage apparatus **10** may include a frame **40** and frame channel **42** that extends across the entire length or any portion of the length of the sleeve **38**. As shown, the frame **40**, and consequently the frame channel **42**, may extend across the entire length of the sleeve **38** to allow the bead **16** to move along the entire length of the finger.

[0039] Referring to FIG. **18**, the finger massage apparatus **10** may comprise a sleeve **38** that is shorter and intended to only cover one of the proximal, medial, or distal phalanges. The shorter sleeve **38** may then be in communication with a shorter frame **40**, and a shorter frame channel **42**. This shorter frame **40** and frame channel **42** provides a shorter path for the movement or translation of the bead **16** to provide a more focused massage experience.

[0040] FIG. **19** provides a depiction of yet another exemplary embodiment of a finger massage apparatus **10** with a sleeve **38**. The sleeve **38** may comprise a first channel **14** and a second channel **15**, each respectively in moveable communication with a bead **16**. As shown, the bead **16** in the

first channel **14** and the bead **16** in the second channel **15** may move independently from one another offering a user a variety of massage or calming options such as movement of the beads **16** in unison, movement of the beads **16** in opposite directions, or any combination thereof.

[0041] Referring to FIGS. **12-19**, the finger massage apparatus **10** with a sleeve **38** may be integrated into one or more fingers of a glove, mitten, or other hand-wearable accessories or clothing.

[0042] FIGS. **20** and **21** provide depictions of exemplary embodiments of a two-ring finger massage apparatus **10** and an exemplary embodiment of a joint. The finger massage apparatus **10** may comprise a first ring **18** and a second ring **19** in communication with the ends of a connecting bar **46** disposed between the first ring **18** and second ring **19**. A bead **16**, comprising a bead channel **48**, may then be in slidable or moveable communication with the connecting bar **46** as the bead **16** may ride along the connecting bar **46** as it passes through the bead channel **48**.

[0043] It should be appreciated that the connecting bar **46** may comprise a flexible or rigid material. Furthermore, the first ring **18**, second ring **19**, and connecting bar **46** may comprise a single molded piece or multiple pieces of the same or dissimilar materials joined together. In certain embodiments, the bead **16** may take on any shape or geometry as desired or required, or as described with reference to other figures herein. Similarly, the connection bar **46** may be straight, curved, or any other shape or geometry as desired or required or as described with reference to other figures herein.

[0044] Referring specifically to FIG. **21**, the connection bar **46** may comprise a joint **50** or other mechanism for allowing flex, displacement, or movement along the length of the connection bar **46**. As shown, the joint **50** may be secured by a pin, rivet, or axle **52** that allows the joint **50** to pivot or rotate about the axle **52**. It should be appreciated that the joint **50** may be any type of joint as is necessary or as desired or required including, but not limited to ball joints, slip joints, leaf joints, or any other type of joint that allows pivoting or rotational movement.

[0045] FIG. **22** provides a depiction of an exemplary embodiment of a clamp style finger massage apparatus **10**. The finger massage apparatus **10** may comprise an upper flange **54** in communication with a lower flange **56** via a joint **58**. A channel **14** in the upper flange **54** may be in moveable communication with a bead **16**, which may slide, rotate, or otherwise translate through the channel **14** to massage a user's finger.

[0046] Still referring to FIG. **22**, the clamp style finger massage apparatus **10** may be adapted for use on the finger tip of a user's finger, or be used across the user's finger at any point along the finger's length. Furthermore, in certain embodiments, the joint **58** may be a solid joint wherein the upper flange **54** and lower flange **56** flex to accept a user's finger. In still further embodiments, the joint **58** may be a hinge, pinned joint, riveted joint, or any other mechanical joint allowing the upper flange **54** and the lower flange **56** to pivot, rotate, or otherwise increase or decrease the spacing between the upper flange **54** and the lower flange **56**.

[0047] FIGS. **23A-23F** provide depictions of several exemplary embodiments of different channel **14** shapes or configurations in a platform **12**. It should be appreciated that any of the shown channel **14** shapes, or any other configuration of channel **14**, may be used in the platform **12**, sleeve **38**, or upper flange **54** of the finger massage apparatuses **10** described herein. For example, as shown in FIG. **23A**, the channel **14** within the platform **12** may be of a straight configuration. The channel **14** may also be a single curve or C-curve as shown in FIG. **23B**. In certain embodiments, the channel **14** may take on an S-curve shape, or a shape with at least one inflection point between concave and convex, as shown in FIG. **23C**. Similarly, the channel **14** may take on a zig-zag shape, or a shape wherein the direction of motion of a bead **16** (not shown) would have abrupt changes of direction as illustrated in FIG. **23D**. It should be appreciated that the various geometries of the channel **14** as displayed in FIGS. **23A-23D** may be adapted for use with any style of bead **16** and may be orientated either parallel with the bones of the user's finger or perpendicular to the bones of the user's finger. Furthermore, the use of different channel **14** geometries allows a user to

experience different sensations or different types of tactile movement which may be adapted for a particular breathing exercise or other calming methods.

[0048] Referring to FIG. 23E, the channel **14**, which is disposed within the platform **12**, but may also be used in conjunction with a sleeve **38** or an upper flange **54** of a finger massage apparatus **10**, may comprise a channel bar **60** and a bead **16** which rides along the channel bar **60**. The channel bar **60** may facilitate different geometries of bead **16**, and allow the bead **16** to both translate along the channel bar **60** and rotate about the channel bar **60** during use.

[0049] Referring specifically to FIG. 23F, the platform **12**, or sleeve **38** or upper flange **54**, of a finger massage apparatus, may have a circular or substantially circular channel **14** and a bead **16** within the circular channel **14**. It should be appreciated that other closed-loop shapes of channel **14** may be used. For example, the channel **14** may be oval, elliptical, square, triangular, or any other shape as desired or required for a particular application.

[0050] FIGS. 24A-24C provide depictions of exemplary embodiments of a platform **12** and channel **14** with detents or other mechanisms for providing intermittent resistance to a bead **16** as it travels along the length of the channel **14**. As shown in FIGS. 24A-24C, the channel **14** may include or comprise one or more obstructions **62**, herein illustrated as nubs **62**, that may provide additional resistance to a bead **16** at certain specific locations along the length of the channel **14**. These locations of additional resistance allow a user to push past the resistance provided by the nubs **62** and give an additional tactile sensation of pushing through the resistance or a “click” feeling to the user during use. It should be appreciated that the nubs **62** may be molded directly into the material of the platform **12**, or they may be molded over of a different material, or may be attached or otherwise affixed to the platform **12**.

[0051] Referring to FIG. 24D, the platform **12** may also incorporate a detent channel **68** within the channel **14** and a ball **64** with a spring **66** to provide the detent function to the finger massage apparatus. When a bead **16** moves along the channel **14** it may encounter the ball **64** and experience resistance to further motion along the channel **14**. When the user applies an additional amount of force, the bead **16** will push the ball **64** against the spring **66**, retracting the ball **64** into the detent channel **68**. When the ball **64** retracts into the detent channel **68**, it will clear the channel **14** and allow the bead **16** to pass.

[0052] FIGS. 25A-25F provide depictions of exemplary embodiments of the engagement of a bead **16**, wheel **74**, barbell **82**, or slider **92** with the channel **14** of a platform **12**. Referring to FIG. 25A, the bead **16** may comprise an upper contact surface **20**, a lower contact surface **21**, and one or more bead channels **22**. The one or more bead channels **22** may be shaped, configured, or otherwise adapted to engaged with or be complementary to the shape of the channel **14**. The engagement of the bead channels **22** with the channel **14** allows the bead **16** to translate or slide along the channel **14** while not being able to escape the channel **14**. In another exemplary embodiment, as shown in FIG. 25B, the bead **16** may comprise one or more bead protrusions **72** which are configured or otherwise adapted to engage one or more channel tracks **70**, which may be located within the channel **14**. It should be appreciated that the bead protrusions **72** may allow the bead to both slide along the channel **14** while also rotating. In certain embodiments, a spherical or substantially spherical bead **16** may engage the channel tracks **70** without the need for a bead protrusion **72** while still constraining the motion of the bead **16** such that it does not fall out of the channel **14**.

[0053] Referring now to FIG. 25C, the finger massage apparatus **10** may include a wheel **74** within the channel **14** in place of the bead **16**. The wheel **74** may itself comprise a disk **76** in communication with an axle **78**. The axle **78** may interface with an axle channel **80** within the channel **14**. The wheel **74** may then translate along the channel **14** and rotate about the axle **78**.

[0054] Referring to FIG. 25D, the channel **14** may have a curved shape that allows a spherical, or substantially spherical, bead **16** to be entrained within the channel **14**. The bead **16** may then rotate about any axis and translate or roll along the length of the channel **14**.

[0055] Referring to FIG. 25E, the finger massage apparatus **10** may include a barbell **82** within the



channel **14**. The barbell **82**, which may comprise one or more balls **84** in communication with a shaft **86**, may pass through the channel **14** such that the balls **84** may extend beyond the surface of the platform **12**. A user may then manipulate the barbell **82** in a similar fashion to a lever in which the lower ball **84** will move in a motion that is opposite to the upper ball **84** as the shaft pivots about in the channel **14**. It should be appreciated that the channel **14** may be an elongated channel to allow the barbell **82** to pivot and translate along the length of the channel **14**. However, in certain embodiments, the channel **14** may be shortened or otherwise constricted such that the barbell **82** may pivot fore and aft or side to side, or pivot in a circular motion but not translate along a length of the platform **12**. As shown, the platform **12** may include additional depressions or clearances on the top surface, bottom surface, or both, to allow a wider range of motion of the barbell **82**.

[0056] Referring to FIG. 25F, the finger massage apparatus **10** may also replace the bead **16** with a slider **92**. The slider **92**, which may comprise an upper portion **94** in communication with a lower portion **96** via a connector **98**. The slider **92** may have a more planar type of geometry, which provides a user with a different tactile sensation during use.

[0057] Still referring to FIG. 25F, the channel **14** may comprise or include a channel track **90** which provides an additional dimension of movement to the slider **92** as it passes along the channel **14**. As shown, the channel track **90** may have a geometry that is not flat or straight, but rather comprises a wave, bumps, or other geometry that tends to move the slider **92** in a vertical direction as it translates through the channel **14**. The vertical movement of the slider **92**, or a bead **16**, wheel **74**, or barbell **82** as illustrated with respect to other embodiments of the finger massage apparatus **10**, may provide changes to pressure to the user's finger to enhance the massaging or calming effects of the finger massage apparatus **10**. It should be appreciated that the connector **98** may take on a geometry to allow the connector **98** to ride along the channel track **90** while permitting the upper portion **94** and lower portion **96** of the slider **92** to extend above and below the surface of the platform **12**, respectively. In certain embodiments, the connector **98** may have a wedged or tapered shape to allow the taper to ride along the channel track **90** while the narrower portion of the connector passes through the channel **14**. Similarly, the connector **98** may comprise a wider section with shoulders that ride along the channel track **90** and a narrower section that may pass completely through the channel **14**. However, any geometry of the connector **98** that allows for engagement of a channel track **90** while passing through the channel **14** may be used.

[0058] FIGS. 26A-26D provide depictions of various exemplary embodiments of a bead **16**. The bead **16** may comprise a spherical or substantially spherical shape as shown in FIG. 26A. The bead **16** may also comprise a shape with a rounded or hemispherical upper contact surface **20** and lower contact surface **21** and one or more bead channels **22**, as shown in FIG. 26B. Referring to FIG. 26C, the bead **16**, which may comprise one or more bead channels **22**, may also comprise a pointed, triangular, or pyramidal shape to the upper contact surface **20**, lower contact surface **21**, or both. FIG. 26D depicts a bead **16** with a flat, or squared upper contact surface **20** and lower contact surface **21** with one or more bead channels **22** disposed therebetween.

[0059] Still referring to FIGS. 26A-D, it should be appreciated that the different shapes of the upper contact surface **20** and lower contact surface **21** may provide a user with different tactile sensations. For example, a rounded contact surface may provide a smoother tactile pressure than a pointed, or pyramidal contact surface, which will provide a more localized pressure to the user. A flat contact surface may provide the most evenly distributed tactile sensation to the user. In certain embodiments, the bead **16** may include an upper contact surface **20** of one configuration with a lower contact surface **21** of a different configuration. In still further embodiments, a bead **16** may be spherical or may have an ellipsoidal or otherwise elongated geometry such that the bead **16** may provide differing amounts of pressure to a user's finger as the bead **16** rotates and engages with the user's finger to a greater or lesser degree.

[0060] FIGS. 27A-27D provide depictions of exemplary embodiments of a wheel **74** for use in a finger massage apparatus **10**. A wheel **74** may comprise an axle channel **80**, which may receive or

otherwise interface with an axle **78**. The wheel **74** may then include a sphere **100**, which may also include an axle channel **80**, as shown in FIG. **27A**. In certain embodiments, the wheel **74** may comprise a disc **76**, as shown in FIG. **27B**. Referring to FIG. **27C**, the disc **76** may comprise one or more disc protrusions **102** disposed about the circumference of the disk **76** to provide additional tactile sensations to a user of the finger massage apparatus **10**. Now referring to FIG. **27D**, the disc **76** of the wheel **74** may include a texture **104** or other geometric features on or about the circumference of the disc **76**.

[0061] FIG. **28** provides a depiction of an exemplary embodiment of a bead **16** for use in a finger massage apparatus **10** which may be split or otherwise separable to facilitate the interchangeability of multiple beads **16** with a single finger massage apparatus **10**. The bead **16**, which may comprise an upper contact surface **20**, a lower contact surface **21**, and one or more bead channels **22**, may further comprise a bead pin **106** in communication with a first half of the bead **16** and a pin recess **108** in communication with a second half of the bead **16**. The bead pin **106** may comprise one or more bead pin ridges **110** or other geometric features. Similarly, the pin recess **108** may comprise complementary ridges **112** or other geometric features. The bead **16** may then be pulled apart to separate the bead **16** into two separate halves. The bead **16** halves may then be extracted from the channel **14** of a finger massage apparatus **10**. A user may then re-connect the two halves of the bead **16** by engaging the bead pin **106** with the pin recess **108**. The bead pin ridges **110** and pin recess ridges **112** allow for a stronger, or more secure connection between the two halves of the bead **16**.

[0062] Still referring to FIG. **28**, it should be appreciated that any number of variations of connection mechanisms may be used to secure the halves of the bead **16** together. For example, the two halves of a bead **16** may be secured together using fasteners, magnets, or other connection means as desired or required for a particular application.

[0063] Referring to FIGS. **25E** and **26A-27D**, it should be appreciated that the bead **16**, wheel **74**, and barbell **82** may take on any number of configurations, geometries, or features when used in the finger massage apparatus **10**. For example, the bead **16**, wheel **74**, and barbell **82** may be made of any number of materials including polymers, elastomers, metals, glass, or any other suitable material. Furthermore, the bead **16**, wheel **74**, and barbell **82** may include or incorporate decorative materials for aesthetic appearance. In certain embodiments, the bead **16**, wheel **74**, and barbell **82** may include stones, including gemstones, crystals, or the like which may provide a pleasing appearance or which may be perceived by a user to enhance the relaxation or healing aspects of the finger massage apparatus. It should also be appreciated that any shape, geometry, or feature discussed in reference to a bead **16**, wheel **74**, or barbell **82** may be adapted for use on another embodiment of a bead **16**, wheel **74**, or barbell **82**. In still further embodiments, the bead **16**, wheel **74**, or barbell **82** may include or comprise different textures or material selections to provide a particular tactile experience. For example, the bead **16**, wheel **74**, or barbell **82** may include knurling, stippling, or other roughening textures, or may include specific metals or plastics to provide more or less heat transfer to the user's finger.

[0064] Referring to FIGS. **1-25E**, it should be appreciated that any features, geometries, or other aspects of the finger massage apparatus **10** described with reference to any one exemplary embodiment may be applied to another exemplary embodiment of the finger massage apparatus **10**. Furthermore, the different geometries and shapes of the channel **14** may be applied to any embodiment, and the connecting bar **46** may also take on a variety of shapes and geometries to provide different paths of travel for the bead **26**. It should be appreciated that any element, part, section, subsection, or component described with reference to any specific embodiment above may be incorporated with, integrated into, or otherwise adapted for use with any other embodiment described herein unless specifically noted otherwise or if it should render the embodiment device non-functional. Furthermore, multiple embodiment devices may be combined, incorporated, or otherwise integrated into one another to construct or develop further embodiments of the invention described herein.

[0065] In summary, while the present invention has been described with respect to specific embodiments, many modifications, variations, alterations, substitutions, and equivalents will be apparent to those skilled in the art. The present invention is not to be limited in scope by any of the specific embodiments described herein. Indeed, various modifications of the present invention, in addition to those described herein, will be apparent to those of skill in the art from the foregoing description and accompanying drawings. Accordingly, the invention is to be considered as limited only by the spirit and scope of the following claims, including all modifications and equivalents.

[0066] It should be appreciated that any element, part, section, subsection, or component described with reference to any specific embodiment above may be incorporated with, integrated into, or otherwise adapted for use with any other embodiment described herein unless specifically noted otherwise or if it should render the embodiment device non-functional. Likewise, any step described with reference to a particular method or process may be integrated, incorporated, or otherwise combined with other methods or processes described herein unless specifically stated otherwise or if it should render the embodiment method nonfunctional. Furthermore, multiple embodiment devices or embodiment methods may be combined, incorporated, or otherwise integrated into one another to construct or develop further embodiments of the invention described herein.

[0067] Still other embodiments will become readily apparent to those skilled in this art from reading the above-recited detailed description and drawings of certain exemplary embodiments. It should be understood that numerous variations, modifications, and additional embodiments are possible, and accordingly, all such variations, modifications, and embodiments are to be regarded as being within the spirit and scope of this application. For example, regardless of the content of any portion (e.g., title, field, background, summary, abstract, drawing figure, etc.) of this application, unless clearly specified to the contrary, there is no requirement for the inclusion in any claim herein or of any application claiming priority hereto of any particular described or illustrated activity or element, any particular sequence of such activities, or any particular interrelationship of such elements. Moreover, any activity can be repeated, any activity can be performed by multiple entities, and/or any element can be duplicated. Further, any activity or element can be excluded, the sequence of activities can vary, and/or the interrelationship of elements can vary. Unless clearly specified to the contrary, there is no requirement for any particular described or illustrated activity or element, any particular sequence or such activities, any particular size, speed, material, dimension or frequency, or any particular interrelationship of such elements. Accordingly, the descriptions and drawings are to be regarded as illustrative in nature, and not as restrictive. Moreover, when any number or range is described herein, unless clearly stated otherwise, that number or range is approximate. When any range is described herein, unless clearly stated otherwise, that range includes all values therein and all sub ranges therein. Any information in any material (e.g., a United States/foreign patent, United States/foreign patent application, book, article, etc.) that has been incorporated by reference herein, is only incorporated by reference to the extent that no conflict exists between such information and the other statements and drawings set forth herein. In the event of such conflict, including a conflict that would render invalid any claim herein or seeking priority hereto, then any such conflicting information in such incorporated by reference material is specifically not incorporated by reference herein.

## Claims

1. A finger massage apparatus comprising: a platform comprising an upper surface and a lower surface; a channel; a bead; and a ring in communication with said lower surface of said platform; wherein said channel is disposed within said platform and said bead is in moveable communication with said channel, and wherein a first portion of said bead extends above said upper surface of said platform and a second portion of said bead extends below said lower surface of said platform.

2. The finger massage apparatus of claim 1, wherein said bead further comprises at least one bead channel; said bead channel adapted to engage said channel of said platform.
3. The finger massage apparatus of claim 1, wherein said bead is removeable and replaceable with at least one second bead.
4. The finger massage apparatus of claim 3, wherein said bead comprises an upper portion and a lower portion; and a bead pin in communication with said upper portion of said bead and a pin recess in communication with said lower portion of said bead; wherein said bead pin may engage said pin recess to secure said upper portion of said bead to said lower portion of said bead.
5. The finger massage apparatus of claim 1, wherein said platform is removably affixed to said ring.
6. The finger massage apparatus of claim 1, wherein said ring comprises a first horn and a second horn; wherein said first horn is in communication with said lower surface of said platform and said second horn is in communication with said lower surface of said platform.
7. The finger massage apparatus of claim 1, wherein said channel comprises a path shape selected from the group consisting of straight, C-curve, S-curve, and zig-zag.
8. The finger massage apparatus of claim 1, wherein at least one of said first portion of said bead and said second portion of said bead comprises a shape selected from the group consisting of spherical, hemispherical, flat, pyramidal, and rounded.
9. The finger massage apparatus of claim 1, further comprising a second channel and a second bead; wherein said second channel is disposed within said platform and said second bead is in moveable communication with said second channel, and wherein a first portion of said second bead extends above said upper surface of said platform and a second portion of said second bead extends below said lower surface of said platform.
10. The finger massage apparatus of claim 1, further comprising a sizing band disposed within said ring; and said sizing band comprising one or more notches and a pull tab; wherein said pull tab of said sizing band may extend through a sizing aperture in said ring to decrease the effective diameter of said ring and adapt said ring to a smaller finger.
11. The finger massage apparatus of claim 1, further comprising one or more detents; said one or more detents disposed within said channel and interfering with said bead to generate a holding force when said bead engages with said one or more detents.
12. A finger massage apparatus comprising: a cylindrical sleeve having an outer surface and an inner surface; and a bead comprising a first portion of said bead and a second portion of said bead in communication with said cylindrical sleeve; wherein said bead is in moveable communication with said cylindrical sleeve and wherein said first portion of said bead extends beyond said outer surface of said cylindrical sleeve and said second portion of said bead extends beyond said inner surface of said cylindrical sleeve.
13. The finger massage apparatus of claim 12, wherein said cylindrical sleeve comprises a flexible material.
14. The finger massage apparatus of claim 12, further comprising a frame, said frame enclosing a channel; wherein said frame is in communication with said cylindrical sleeve, and said bead is in moveable communication with said channel.
15. The finger massage apparatus of claim 12, further comprising a sheath; wherein said sheath is in communication with said cylindrical sleeve and encloses a space; and wherein said bead is moveably disposed within said space defined by said sleeve.
16. The finger massage apparatus of claim 12, further comprising a channel in communication with said cylindrical sleeve; wherein said channel is disposed along an axis of said cylindrical sleeve.
17. The finger massage apparatus of claim 12, further comprising a channel in communication with said cylindrical sleeve; wherein said channel is disposed along a circumference of said cylindrical sleeve.
18. A finger massage apparatus comprising: a first ring; a second ring; a connection bar having a

first end and a second end; and a bead; wherein said first ring is in communication with said first end of said connection bar, said second ring is in communication with said second end of said connection bar, and said bead is in slidable communication with said connection bar and disposed between said first ring and said second ring.

**19.** The finger massage apparatus of claim 18, wherein said connection bar comprises a flexible material.

**20.** The finger massage apparatus of claim 18, wherein said connection bar comprises a first portion and a second portion; and wherein said first portion of said connection bar is pivotably affixed to said second portion of said connection bar through a joint.

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